

Rudder Analytics Round 2: Question Bank (Questions Only)

For Karan | Practice set for Fri 9 Oct 2026 | No solutions, attempt each in Python (or SQL where marked)

How to use: Start with the section marked MUST DO (1 to 4 and 12). Time yourself: about 10 min per easy problem. Tick off what you finish. Then try the Mock Tests in section 14 as a final check.

1. Strings (MUST DO)

1. Reverse a given string.
2. Check whether a string is a palindrome (ignore case, spaces and punctuation).
3. Count the vowels and consonants in a string.
4. Check whether two strings are anagrams.
5. Print the frequency of every character in a string.
6. Find the first non-repeating character in a string (print None if there is none).
7. Find the first repeating character in a string.
8. Reverse the order of words in a sentence.
9. Reverse every word in a sentence but keep the word order.
10. Compress a string: 'aaabbc' becomes 'a3b2c1'.
11. Decompress a string: 'a3b2c1' becomes 'aaabbc'.
12. Remove all duplicate characters from a string, keeping the first occurrence.
13. Find the longest word in a sentence.
14. Count the number of words in a sentence that may contain extra spaces.
15. Check whether a string is a pangram.
16. Swap the case of every letter in a string without using swapcase().
17. Capitalise the first letter of every word without using title().
18. Find the most frequent character in a string.
19. Find the longest common prefix among a list of strings.
20. Remove all spaces and special characters from a string.

21. Count the uppercase letters, lowercase letters, digits and special characters in a string.
22. Check whether a string contains only digits (without using `isdigit()`).
23. Replace every vowel in a string with '*'.
24. Check whether a string is a rotation of another string ('abcd' and 'cdab').
25. Find all substrings of a given string.
26. Print all characters at even indexes of a string.
27. Find the length of the longest substring without repeating characters.
28. Check whether brackets in a string are balanced: (), [], {}.
29. Convert a string to its ASCII codes and back.
30. Given a sentence, print each word with its length.

2. Arrays / Lists (MUST DO)

1. Find the largest and smallest element in a list.
2. Find the second largest element in a list (handle duplicates).
3. Find the third largest element in a list.
4. Sort a list in ascending order without using `sort()` or `sorted()`.
5. Sort a list in descending order using a built-in method.
6. Remove duplicates from a list while preserving order.
7. Rotate a list to the right by k positions.
8. Rotate a list to the left by k positions.
9. Find two numbers in a list that add up to a target and print their indexes.
10. Find the missing number in a list containing 1 to n with one number missing.
11. Move all zeros in a list to the end, keeping the order of other elements.
12. Find the maximum subarray sum (Kadane).
13. Merge two sorted lists into one sorted list.
14. Search an element in a sorted list using binary search.
15. Find all duplicate elements in a list.
16. Find the element that appears most often in a list.
17. Count the pairs in a list whose sum equals k.
18. Print the elements at even indexes and odd indexes separately.
19. Reverse a list in place without using `reverse()` or slicing.

20. Check whether a list is sorted in ascending order.
21. Find the sum and average of all elements of a list.
22. Separate even and odd numbers into two lists.
23. Find the common elements between two lists.
24. Find the elements that are in one list but not in the other.
25. Find the union and intersection of two lists.
26. Print the cumulative (running) sum of a list.
27. Find the longest increasing run of consecutive elements.
28. Find the leaders in a list (elements greater than everything to their right).
29. Find the equilibrium index (left sum equals right sum).
30. Segregate 0s and 1s in a list.
31. Sort a list containing only 0, 1 and 2 without using sort().
32. Find the product of all elements except the current one (without division).
33. Find the majority element (appears more than $n/2$ times).
34. Check whether a list is a palindrome.
35. Given a list of numbers, print the sum of squares of the even numbers only.
36. Find the k-th smallest element in a list.
37. Insert an element at a given position and delete an element by value.
38. Find the length of the longest consecutive sequence in an unsorted list.
39. Find the difference between the largest and smallest element.
40. Given a list of marks, print the students above the average.

3. Numbers and Maths (MUST DO)

1. Check whether a number is prime.
2. Print all prime numbers up to n.
3. Print the first n terms of the Fibonacci series.
4. Find the factorial of a number (iterative and recursive).
5. Find the GCD and LCM of two numbers.
6. Find the sum of digits of a number.
7. Reverse a number.
8. Check whether a number is a palindrome.
9. Check whether a number is an Armstrong number.

10. Print FizzBuzz from 1 to n.
11. Count the number of digits in a number.
12. Check whether a number is a perfect number.
13. Check whether a number is a perfect square without using sqrt.
14. Convert a decimal number to binary without using bin().
15. Convert a binary number to decimal.
16. Find the power of a number without using ** or pow().
17. Check whether a number is even or odd without using %.
18. Print the multiplication table of a number.
19. Find the sum of the first n natural numbers using a loop and using a formula.
20. Find all factors of a number.
21. Find the prime factorisation of a number.
22. Check whether a number is a strong number (sum of digit factorials equals the number).
23. Print all Armstrong numbers in a range.
24. Swap two numbers without using a third variable.
25. Find the largest of three numbers.
26. Check whether a year is a leap year.
27. Compute simple interest and compound interest from principal, rate and time.
28. Calculate the average of n numbers entered by the user.
29. Find the number of trailing zeros in n factorial.
30. Print the Collatz sequence of a number.

4. Linked List (MUST DO)

Assume a Node class with data and next.

1. Create a linked list from a Python list and print it.
2. Insert a node at the beginning, at the end and at a given position.
3. Delete the first node, the last node and a node with a given value.
4. Find the length of a linked list.
5. Search for a value in a linked list.
6. Reverse a linked list.
7. Find the middle node of a linked list.

8. Detect whether a linked list has a cycle.
9. Merge two sorted linked lists.
10. Remove the n-th node from the end.
11. Remove duplicates from a sorted linked list.
12. Check whether a linked list is a palindrome.
13. Find the n-th node from the end.
14. Print the linked list in reverse order without modifying it.
15. Add two numbers represented as linked lists (digits stored in reverse order).
16. Find the intersection node of two linked lists.
17. Swap adjacent nodes in a linked list.
18. Count the occurrences of a value in a linked list.

5. Pattern Printing

1. Right-angled triangle of stars.
2. Inverted right-angled triangle of stars.
3. Pyramid of stars.
4. Inverted pyramid of stars.
5. Diamond of stars.
6. Number triangle (1 / 1 2 / 1 2 3 ...).
7. Floyd's triangle.
8. Pascal's triangle for n rows.
9. Hollow square of stars.
10. Alphabet triangle (A / A B / A B C ...).

6. Matrix (2D lists)

1. Print a matrix row by row.
2. Find the transpose of a matrix.
3. Add two matrices.
4. Multiply two matrices.
5. Find the sum of the main diagonal and the secondary diagonal.
6. Find the row-wise and column-wise sums.

7. Print a matrix in spiral order.
8. Rotate a square matrix by 90 degrees.
9. Check whether a matrix is symmetric.
10. Find the largest element in each row.

7. Dictionary, Set and Counter

1. Count the frequency of each word in a sentence.
2. Find the word that occurs most often in a paragraph.
3. Group words that are anagrams of each other.
4. Find the first non-repeating element in a list using a dictionary.
5. Merge two dictionaries and sum the values of common keys.
6. Sort a dictionary by value in descending order.
7. Invert a dictionary (swap keys and values).
8. Find the elements present in the first list but not in the second using sets.
9. Given student names and marks, print the topper and the average.
10. Check whether all elements of a list are unique.
11. Find the pair of elements with the given difference k.
12. Count how many times each vowel appears in a string.

8. Recursion, Sorting and Searching

1. Sum of digits using recursion.
2. Factorial using recursion.
3. Fibonacci using recursion.
4. Power of a number using recursion.
5. Reverse a string using recursion.
6. Check a palindrome using recursion.
7. Implement bubble sort, selection sort and insertion sort.
8. Implement merge sort.
9. Implement linear search and binary search.
10. Tower of Hanoi for n disks (print the moves).
11. Generate all permutations of a short string.

12. Generate all subsets of a small list.

9. File Handling and Basic I/O

1. Read a text file and count its lines, words and characters.
2. Write a list of items into a file, one per line.
3. Read numbers from a file and print their sum.
4. Copy the contents of one file into another.
5. Take n integers as input on one line and print their sum.
6. Take n lines of input and print them in reverse order.
7. Read a name and age from the user and print a formatted sentence.
8. Read comma-separated values and convert them to a list of integers.

10. SQL (write queries)

Tables: employees(id, name, dept_id, salary, manager_id, join_date), departments(id, dept_name), orders(order_id, customer_id, amount, order_date), customers(customer_id, name, city).

1. Display all employees.
2. Display the names and salaries of employees earning more than 50000.
3. Join employees and departments to show employee name and department name.
4. Show all employees even if they have no department (LEFT JOIN).
5. Count the number of employees in each department.
6. Show departments having more than 3 employees.
7. Find the highest salary and the second highest salary.
8. Find the top 3 highest paid employees.
9. Find duplicate employee names.
10. Find employees earning more than the average salary.
11. Find the department with the highest average salary.
12. Show each employee along with their manager's name (self join).
13. Find customers who have placed no orders.
14. Find the total order amount per customer, highest first.
15. Find customers from a given city with total orders above 10000.

16. Find employees who joined in the last 6 months.
17. Delete duplicate rows keeping one copy.
18. Find the nth highest salary.
19. Show the number of orders per month.
20. Explain the difference between WHERE and HAVING, and INNER vs LEFT vs RIGHT vs FULL JOIN.

11. Pandas and Data Handling

1. Load a CSV file and show the first five rows, the shape and the column types.
2. Select rows where a column value is above a threshold.
3. Find the number of missing values in each column.
4. Fill missing numeric values with the column mean.
5. Drop duplicate rows.
6. Group by a column and find the mean, sum and count of another column.
7. Sort a DataFrame by a column in descending order.
8. Create a new column from two existing columns.
9. Merge two DataFrames on a common key.
10. Find the top 5 most frequent values of a column.
11. Convert a string column to datetime and extract the month.
12. Compute the correlation between two numeric columns.
13. Use NumPy to compute the mean, median and standard deviation of an array.
14. Create a 3x3 matrix with NumPy and find its transpose and the sum of each row.

12. Predict the Output (MUST SKIM)

Write down the output of each snippet before running it.

1. `print(7 / 2, 7 // 2, -7 // 2, 7 % 3)`
2. `print(0.1 + 0.2 == 0.3)`
3. `a = [1, 2]; b = a; b.append(3); print(a)`
4. `a = [1, 2]; b = a.copy(); b.append(3); print(a)`
5. `def f(x, lst=[]): lst.append(x); return lst` then `print(f(1), f(2))`
6. `print('hello'[10:])` and then `print('hello'[10])`

7. `s = 'abc'; s[0] = 'x'`
8. `print(bool([]), bool(' '), bool(0), bool('0'))`
9. `print(5 == 5.0, 5 is 5.0)`
10. `grid = [[0] * 3] * 3; grid[0][0] = 1; print(grid)`
11. `for i in range(5): pass then print(i)`
12. `print(sorted('bca'), ".join(sorted('bca')))`
13. `print(type(input()))` when the user types 5
14. `x = 10; def f(): x = 20; f(); print(x)`
15. `print([i * 2 for i in range(5) if i % 2 == 0])`
16. `print('a' + 'b' * 3, 'ab' * 0)`
17. `t = (1, 2, 3); t[0] = 5`
18. `d = {'a': 1, 'b': 2}; print(d.get('c', 0), len(d))`
19. `print(list(range(10, 0, -3)))`
20. `s = {1, 2, 2, 3}; print(len(s), s)`
21. `print(max('apple', 'banana'), min([3, 1, 2]))`
22. `i = 0; while i < 3: print(i); i += 1; else: print('done')`
23. `print(2 ** 3 ** 2)`
24. `a, b = 1, 2; a, b = b, a + b; print(a, b)`
25. `print(int('12') + int(3.9), str(5) + '5')`

13. Core Concepts (be ready for a quick theory question)

1. Difference between a list, tuple, set and dictionary.
2. Mutable vs immutable types with examples.
3. What is the difference between `==` and `is`?
4. What are `*args` and `**kwargs`?
5. What is a lambda function? Write one that squares a number.
6. What is a list comprehension? Convert a for loop to one.
7. What is the difference between a shallow copy and a deep copy?
8. Explain `try`, `except`, `else` and `finally`.
9. What is a class, an object, inheritance and polymorphism? Write a small class with a method.
10. What is the time complexity of searching in a list vs a set?

11. What is the difference between a stack and a queue? Implement each with a list.
12. What is a generator and how does yield work?
13. What is the difference between append() and extend()?
14. What is a decorator (basic idea)?

14. Timed Mock Tests (final check, 90 min each)

Mock Test 1 (2 questions)

1. Read a sentence and print the words sorted by length (longest first), along with each word's length. Words of equal length keep their original order.
2. Read n integers and an integer k. Print the k-th largest distinct number, or -1 if fewer than k distinct numbers exist.

Mock Test 2 (2 questions)

1. Build a linked list from n integers, reverse it, then remove all nodes with a given value and print the final list.
2. Read a string and print the character that appears most often along with its count. On ties, print the one that appears first in the string.

Mock Test 3 (2 questions)

1. Given a list of integers, print the length of the longest subarray with sum equal to zero.
2. Given two strings, print whether they are anagrams, and if not, print the characters that differ along with their counts.

Mock Test 4 (2 questions)

1. Given a matrix of n by m numbers, print the sum of each row, the sum of each column, and the position (row, column) of the largest element.
2. Given a string of brackets, print whether it is balanced; if not, print the index of the first unmatched bracket.

Mock Test 5 (2 questions)

1. Given employee names and salaries as input lines, print the employee with the second highest salary and the average salary rounded to 2 decimal places.
2. Given a list of numbers, print all pairs that sum to a target without repeating the same pair twice.

15. Final Checklist Before You Sleep

1. Can you write from memory: string reverse, palindrome, anagram, second largest, two sum, Kadane, rotate list, binary search?
2. Can you build a linked list, reverse it and find the middle without looking?
3. Can you read input with `input().split()` and print in the exact required format?
4. Have you done at least 10 problems today by typing them yourself, not just reading?
5. Slot time checked in email (including Spam), webcam and charger ready, Incognito window planned?